Computer Networks

CSE Semester 5 — Academic Notes

Synthetic academic knowledge-base document created for CampusAI RAG evaluation and demonstration.

Unit 1 — Network Fundamentals

- A computer network is an interconnected collection of devices that exchange data using communication protocols.
- Network performance is commonly discussed using bandwidth, throughput, latency, jitter, and packet loss.
- Network topologies include bus, star, ring, mesh, and hybrid forms.
- The OSI reference model contains seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application.
- The TCP/IP architecture commonly uses Link, Internet, Transport, and Application layers.

Unit 2 — Data Link and Ethernet

- The Data Link layer provides node-to-node delivery and handles framing, error detection, and media access control.
- Ethernet is a widely used LAN technology. A MAC address identifies a network interface at the data-link layer.
- Error detection techniques include parity, checksums, and cyclic redundancy check (CRC).
- Switches forward frames using MAC address tables and generally operate at the Data Link layer.
- Collision handling in traditional shared Ethernet used CSMA/CD; modern switched full-duplex Ethernet normally avoids collisions.

Unit 3 — Network Layer and Routing

- The Network layer provides logical addressing and routing between networks.
- IPv4 uses 32-bit addresses. Subnetting divides an address space into smaller logical networks using a subnet mask or prefix length.
- Routers forward packets based on routing tables.
- Routing protocols can be classified as distance-vector or link-state. RIP is a distance-vector protocol; OSPF is a link-state protocol.
- IPv6 uses 128-bit addresses and provides a much larger address space than IPv4.

Unit 4 — Transport Layer

- TCP provides connection-oriented, reliable, ordered byte-stream delivery. UDP provides connectionless datagram delivery with lower protocol overhead.
- TCP reliability uses sequence numbers, acknowledgements, retransmissions, and flow control.
- TCP congestion control adjusts the sending rate according to network conditions.
- Flow control prevents a sender from overwhelming the receiver. Congestion control addresses overload within the network.
- Ports identify application endpoints. A socket is commonly described using an IP address and port number.

Unit 5 — Application Layer

- DNS maps domain names to IP addresses and other resource records.
- HTTP is an application-layer protocol used for web communication. HTTPS adds security through TLS.
- SMTP is used for email transfer. POP3 and IMAP are protocols used for retrieving email.
- DHCP dynamically provides network configuration such as IP address, subnet mask, gateway, and DNS server.
- RESTful web services commonly use HTTP methods such as GET, POST, PUT/PATCH, and DELETE.

Quick Revision Checklist

- Know the key definitions and distinguish closely related concepts.
- Practice explaining each concept in your own words before using it in an exam.
- For numerical or algorithmic topics, work through at least one example by hand.
- For database and software-engineering topics, connect definitions to practical examples.

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